

WEEK 1:

- **Economics:** Science that deals with allocation of scarce resources to meet human wants.
- **Needs:** essential for survival and well-being
- **Wants:** once the basic needs are met, our aspirations goes up → wants are known to enhance the quality of life.
- Difference of wants with desire: in terms of affordability
- **Resources** → tend to be scarce in nature

Goods:

1. Economic goods: goods which are scarce in nature
2. Free goods: goods where there is enough supply to meet demands

Production:

- Any activity carried out that satisfies a human want.
- process that converts raw materials into usable goods/services. It provides value-addition, and as a result has a utility value.
 - Land
 - Labour
 - Capital
 - Enterprise → combine land + labour + capital to produce goods and services

Micro	Macro
Individual consumers/businesses and the financial decisions they make	Economy as a whole
Smaller parts of the economy	Larger economy
Involves looking at the society from bottom up	Involves looking at the society from top down

Demand:

- Desire backed by ability and willingness to pay.
- created by consumption, not by production or any other activity

Scarcity:

- Short supply of resources.

Causes of scarcity:

1. **Demand induced scarcity** : supply is same, demand increases
2. **Supply induced scarcity** : demand is same, supply reduces
3. **Structural scarcity** : some people have less access to resources in comparison to others

Utility

- Refers to the satisfaction consumer gets upon consumption of goods/services
- Influences economic decisions

Marginality:

- Economic decisions made bit by bit, not all in one go.
- **Marginal utility** → incremental increase in utility that is derived from the consumption of one additional unit. (eg: satisfaction increases by a fraction)

Law of diminishing marginal utility:

- As consumption increases, marginal utility decreases. (satisfaction reduces)
- Assumptions:
 - Items that are consumed are identical
 - Units/ items are consumed quickly and there are few breaks in between
 - Consumer is in a position to make rational decisions wrt consumption
 - Items can be measured
 - Consumers taste remains constant
 - Units consumed are neither too big nor too small

Economy classes:

1. Capitalist:

- **Govt plays a minimal role**

2. Socialist

- Rule based system
- Rules decided by central authorities

3. Mixed economy

- mix of capitalist and socialist market
- Private businesses operate freely, but govt has some regulations

4. Traditional

- Very little govt or market influence

5. Open economy:

- Anyone can invest/trade

6. Closed economy:

- No exchange out of the country

Rationality:

- People make the cost-benefit analysis/assessments. → this is specific to every person
- They make decisions when $\text{benefits} > \text{costs}$ (benefits outweighs costs), and may not make decisions when $\text{benefits} < \text{costs}$

Opportunity Costs:

When the consumers make a choice, they have to give up/reject another choice → the cost associated with such lost opportunities is known as “opportunity costs”

- **Explicit Opportunity cost:** direct out of pocket payment for a resource (actual cash flow)

- **Implicit Opportunity cost:** a hidden non cash opportunity cost – when you chose one option over an other (value given up)

Specialization:

- In most countries, people tend to earn more money by specializing in particular tasks. → known as division of labour
- Economic sectors:
 1. Primary → extract raw materials from the earth (fishing/farming/mining)
 2. Secondary → conversion of raw materials into finished products (electricity/fuel)
 3. Tertiary → service sector
 4. Quaternary → scientific R&D, education, public sector

The below passage is common for Questions 7 to 8

The conceptual and numerical problems are based on the hypothetical case. Read the case carefully and based on the Economics concepts taught in class answer the following questions:

Case Scenario-1:

A GreenFleet001 startup in Chennai manages a delivery fleet for online food delivery partners using two types of vehicles: **Electric 2- Wheelers (E2)** and **Electric 3-Wheelers (E3)**. The company's fleet operations are subject to a strict daily battery-charging cap of ₹20,000.

Daily Budget Constraint Equation: $100E_2 + 400E_3 = 20,000$

Where:

Charging costs associated with vehicles:

₹100 is the daily charging cost per **Electric 2- Wheelers (E2)**.

₹400 is the daily charging cost per **Electric 3-Wheelers (E3)**.

Profit associated with the Vehicles:

Electric 2- Wheelers Cycles (E2) generates ₹500 in daily net profit.

Electric 2- Scooty/Bikes (E3) generates ₹1,500 in daily net profit.

Due to a new, mandatory Business to Business (B2B) contract with a local e-commerce delivery based company, the startup **must deploy exactly 10 Electric 3-Wheelers (E3 = 10)** every day. To maximize overall company returns, the fleet manager wants to optimize the remaining charging budget to deploy as many Electric 2-Wheelers as possible.

$$100E_2 + 400E_3 = 20,000 \quad E_3 = 10$$

$$\Rightarrow 100E_2 + 400(10) = 20,000 \quad \rightarrow E_2 = \frac{20,000 - 4,000}{100} = 160$$

$$\text{Max profit: } 500E_2 + 1500E_3 = 160 \times 500 + 1500 \times 10 \\ = 80,000 + 15,000 = 95,000$$